MAT-121
Differential Calculus
By Das and Mukherjee

51st Edition

Chapter: Limits

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Chapter-01: Limits

❖ **Limit of a function: $\{\underset{x \to a}{lt} f(x)\}$**

When x approaches a constant quantity 'a' from either side (but $x \neq a$), if there exists a definite finite number l towards which $f(x)$ approaches, such that the numerical difference of $f(x)$ and l can be made as small as we please (i.e., less than any pre-assigned positive quantity, however small) by taking x sufficiently close to a, then l is defined as the limit of $f(x)$ as x tends to a.

This is symbolically written as $\underset{x \to a}{lt} f(x) = l$.

❖ Limit of a function: Cont.

- **Mathematically,** $\underset{x \to a}{lt} f(x) = l$, provided, given any pre-assigned positive quantity ε, however small; we can determine another pre-assigned positive quantity δ (depending on ε) such that $|f(x) - l| <$ ε for all values of *x* satisfying $0 < |x - a| \ll \delta$, i.e.,

whenever $a - \delta \ll x \ll a + \delta$, **but *x* ≠ *a*.**

❖ Left Hand Limit: { $\underset{x \to a^-}{lt} f(x)$}

- $\underset{x \to a^-}{lt} f(x) = l_1$, provided, given any pre-assigned positive quantity ε, however small; we can determine another pre-assigned positive quantity δ (depending on ε) so that $|f(x) - l_1| < \varepsilon$

  Whenever $0 < a - x < \delta$,

  i.e., $a - \delta \ll x < a$.

❖ Right Hand Limit: { $\underset{x \to a^+}{lt} f(x)$}

- $\underset{x \to a^+}{lt} f(x) = l_2$, provided, given any pre-assigned positive quantity ε, however small; we can determine another pre-assigned positive quantity δ (depending on ε) so that $|f(x) - l_2| < ε$

  Whenever $0 < x - a \ll δ$,

  i.e., $a < x \ll a + δ$.

Ex.1 If $f(x) = \frac{|x|}{x}$ then does $\lim_{x\to 0} f(x)$ exist?

Solution:

We have $$f(x) = \begin{cases} \frac{x}{x}, \text{ when } x > 0 \\ \frac{-x}{x}, \text{ when } x < 0 \end{cases}$$

L.H.L = $\underset{x\to 0^-}{lt} f(x) = \underset{x\to 0^-}{lt} \frac{-x}{x} = \underset{x\to 0^-}{lt} (-1) = -1$

R.H.L = $\underset{x\to 0^+}{lt} f(x) = \underset{x\to 0^+}{lt} \frac{x}{x} = \underset{x\to 0^+}{lt} (1) = 1$

- Since L.H.L $\neq$ R.H.L
- So $\lim_{x\to 0} f(x)$ does not exists.

Ex.2 If $f(x) = \begin{cases} 1+2x, when \ -\frac{1}{2} \le x < 0 \\ 1-2x, when \ 0 \le x < \frac{1}{2} \\ -1+2x, when \ x \ge \frac{1}{2} \end{cases}$; then find $\lim_{x \to \frac{1}{2}} f(x)$.

Solution:

We have $f(x) = \begin{cases} 1+2x, when \ -\frac{1}{2} \le x < 0 \\ 1-2x, when \ 0 \le x < \frac{1}{2} \\ -1+2x, when \ x \ge \frac{1}{2} \end{cases}$;

L.H.L = $\underset{x \to \left(\frac{1}{2}\right)^-}{lt} f(x) = \underset{x \to \left(\frac{1}{2}\right)^-}{lt} (1-2x) = 1 - 2.\frac{1}{2} = 1 - 1 = 0$

R.H.L = $\underset{x \to \left(\frac{1}{2}\right)^+}{lt} f(x) = \underset{x \to \left(\frac{1}{2}\right)^+}{lt} (-1+2x) = -1 + 2.\frac{1}{2} = -1 + 1 = 0$

Since L.H.L = R.H.L

So, $\lim_{x \to \frac{1}{2}} f(x) = 0$. (**Answer**).

Ex.3 If $f(x) = \begin{cases} \tan\left(\frac{x}{2}\right), when\ x < \frac{\pi}{2} \\ 3 - \frac{\pi}{2}, when\ x = \frac{\pi}{2} \\ \frac{x^3 - \frac{\pi^3}{8}}{x - \frac{\pi}{2}}, when\ x > \frac{\pi}{2} \end{cases}$; then find $\lim_{x \to \frac{\pi}{2}} f(x)$ if possible.

Solution: We have $f(x) = \begin{cases} \tan\left(\frac{x}{2}\right), when\ x < \frac{\pi}{2} \\ 3 - \frac{\pi}{2}, when\ x = \frac{\pi}{2} \\ \frac{x^3 - \frac{\pi^3}{8}}{x - \frac{\pi}{2}}, when\ x > \frac{\pi}{2} \end{cases}$

L.H.L = $\underset{x \to \left(\frac{\pi}{2}\right)^-}{lt} f(x) = \underset{x \to \left(\frac{\pi}{2}\right)^-}{lt} \tan\left(\frac{x}{2}\right) = \tan\left(\frac{\frac{\pi}{2}}{2}\right) = \tan\left(\frac{\pi}{4}\right) = 1$

R.H.L = $\underset{x \to \left(\frac{\pi}{2}\right)^+}{lt} f(x) = \underset{x \to \left(\frac{\pi}{2}\right)^+}{lt} \frac{x^3 - \frac{\pi^3}{8}}{x - \frac{\pi}{2}} = \underset{x \to \left(\frac{\pi}{2}\right)^+}{lt} \frac{\left(x - \frac{\pi}{2}\right)\left(x^2 + x.\frac{\pi}{2} + \frac{\pi^2}{4}\right)}{\left(x - \frac{\pi}{2}\right)}$

Ex.3 Cont.

$$\text{R.H.L} = \lim_{x \to \left(\frac{\pi}{2}\right)^+} \left(x^2 + x.\frac{\pi}{2} + \frac{\pi^2}{4} \right) = \left(\frac{\pi}{2}\right)^2 + \frac{\pi}{2}.\frac{\pi}{2} + \frac{\pi^2}{4}$$

$$= \frac{\pi^2}{4} + \frac{\pi^2}{4} + \frac{\pi^2}{4}$$

$$= \frac{3\pi^2}{4}$$

Since $\text{L.H.L} \neq \text{R.H.L}$

So $\lim_{x \to \frac{\pi}{2}} f(x)$ is not possible.

Ex.5 If $f(x) = \begin{cases} x, when\ x > 0 \\ 0, when\ x = 0 \\ -x, when\ x < 0 \end{cases}$; then find the value of $\lim_{x \to 0} f(x)$.

Solution: We have $f(x) = \begin{cases} x, when\ x > 0 \\ 0, when\ x = 0 \\ -x, when\ x < 0 \end{cases}$

$$L.H.L = \underset{x \to 0^-}{lt} f(x) = \underset{x \to 0^-}{lt} -x = (-0) = 0$$

$$R.H.L = \underset{x \to 0^+}{lt} f(x) = \underset{x \to 0^+}{lt} x = 0$$

Since L.H.L = R.H.L

So $\lim_{x \to 0} f(x) = 0$.(**Answer**).

<u>Ex.6</u> A function $\varphi(x)$ is defined as follows:

$$\varphi(x) = \begin{cases} x^2, & for\ x < 1 \\ 2.5, & for\ x = 1 \\ x^2 + 2, & for\ x > 1 \end{cases}$$
; then find the value of $\lim_{x \to 1} \varphi(x)$.

Solution:

We have $\varphi(x) = \begin{cases} x^2, & for\ x < 1 \\ 2.5, & for\ x = 1 \\ x^2 + 2, & for\ x > 1 \end{cases}$

L.H.L $= \lim_{x \to 1^-} \varphi(x) = \lim_{x \to 1^-} x^2 = 1^2 = 1$

R.H.L $= \lim_{x \to 1^+} \varphi(x) = \lim_{x \to 1^+} (x^2 + 2) = 1^2 + 2 = 3$

Since L.H.L $\neq$ R.H.L

So, $\lim_{x \to 1} \varphi(x)$ does not exists.

Ex.7 If $f(x) = \begin{cases} x \ if \ 0 < x < 1 \\ 2 - x \ \ if \ 1 \le x \le 2 \\ x - \frac{x^2}{2} \ \ if \ x \ge 2 \end{cases}$; does limit exists at $x = 1$ and $x = 2$?

Solution: We have $f(x) = \begin{cases} x \ if \ 0 < x < 1 \\ 2 - x \ \ if \ 1 \le x \le 2 \\ x - \frac{x^2}{2} \ \ if \ x \ge 2 \end{cases}$

- **When $x = 1$**

L.H.L $= \lim\limits_{x \to 1^-} f(x) = \lim\limits_{x \to 1^-} x = 1$

R.H.L $= \lim\limits_{x \to 1^+} f(x) = \lim\limits_{x \to 1^+} (2 - x) = 2 - 1 = 1$

Since L.H.L = R.H.L

So, $\lim\limits_{x \to 1} f(x) = 1$

Ex.7 Cont.

- **When $x = 2$**

$$\text{L.H.L} = \lim_{x \to 2^-} f(x) = \lim_{x \to 2^-} (2 - x) = 2 - 2 = 0$$

$$\text{R.H.L} = \lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (x - \frac{x^2}{2}) = 2 - \frac{2^2}{2} = 2 - 2 = 0$$

Since L.H.L = R.H.L

So, $\lim_{x \to 2} f(x) = 0$

Answer: $\lim_{x \to 1} f(x) = 1$ **&** $\lim_{x \to 2} f(x) = 0$

Theorem: Prove that $\underset{x \to a}{lt} \frac{x^n - a^n}{x - a} = na^{n-1}$ for all rational values of *n* provided *a* is a positive.

- **Proof**: **Case-I**: When *n* is a positive integer.

Then $\underset{x \to a}{lt} \frac{x^n - a^n}{x - a} = \underset{x \to a}{lt} \frac{(x-a)\left(x^{n-1} + x^{n-2}.a + \ldots + a^{n-1}\right)}{(x-a)}$

$$= \underset{x \to a}{lt} \{x^{n-1} + x^{n-2}.a + \ldots + a^{n-1}\}$$

$$= a^{n-1} + a^{n-2}.a + \ldots + a^{n-1}$$

$$= a^{n-1} + a^{n-1} + \ldots + a^{n-1}$$

$$= na^{n-1}$$

Thus $\underset{x \to a}{lt} \frac{x^n - a^n}{x - a} = na^{n-1}$

Theorem: Cont.

- **Case-II**: When n is negative integer.

  Let n = -m, m being a positive integer and $a \neq 0$.

$$\therefore \underset{x \to a}{lt} \frac{x^n - a^n}{x - a} = \underset{x \to a}{lt} \frac{x^{-m} - a^{-m}}{x - a}$$

$$= \underset{x \to a}{lt} \left(\frac{1}{x^m} - \frac{1}{a^m} \right) \left(\frac{1}{x - a} \right)$$

$$= \underset{x \to a}{lt} \left(\frac{a^m - x^m}{x^m a^m} \right) \left(\frac{1}{x - a} \right)$$

$$= \underset{x \to a}{lt} \left\{ \left(-\frac{1}{x^m a^m} \right) \left(\frac{x^m - a^m}{x - a} \right) \right\}$$

$$= -\frac{1}{a^m} \underset{x \to a}{lt} \left(\frac{1}{x^m} \right) \underset{x \to a}{lt} \left(\frac{x^m - a^m}{x - a} \right)$$

Theorem: Cont. **Case-II**:

- **Case-II**

$$or,\ \underset{x\to a}{lt}\frac{x^n - a^n}{x - a} = -\frac{1}{a^m}.\frac{1}{a^m}.ma^{m-1}$$

$$= -\frac{1}{a^{2m}}.ma^{m-1}$$

$$= -ma^{m-1-2m}$$

$$= -ma^{-m-1}$$

$$= na^{n-1}$$ [Since n=-m]

So, $\underset{x\to a}{lt}\frac{x^n-a^n}{x-a} = na^{n-1}$

Hence $\underset{x\to a}{lt}\frac{x^n-a^n}{x-a} = na^{n-1}$